

Annotation-Aware Indexing: Efficient Query Processing over Machine-Labelled Text Collections

Project outline for the PhD position in Developing Efficient Search Engines (NTNU, Jobbnorge 307690) · Asad Ikram

1. Problem and motivation

Large text collections are no longer searched only by their words. Social media archives, news collections and review corpora are now routinely passed through classifiers that attach labels and scores to every document: sentiment, emotion, topic, language, toxicity, each with a confidence value. The questions analysts then ask combine all of these: *posts mentioning inflation, in Urdu or English, labelled negative with confidence above 0.8, from the week of a budget announcement*. Two features make this hard for a search engine. First, the query mixes keyword relevance with categorical filters and numeric ranges over annotations. Second, the annotations are not stable: when a classifier is retrained, a new model version relabels a large part of the corpus, and an index built on the old labels silently answers a different question.

I meet this daily. At ArtemisAI six fine-tuned classifiers annotate more than three million comments, and the models have gone through several versions. In my MSc research on about 279,000 posts, the same documents received different labels from a lexicon, a BiLSTM and a fine-tuned BERT. The Databases and Search group has already built infrastructure for search over semantically annotated text (GYANI, CIKM 2018) and for range search over temporal and numerical expressions (ICTIR 2024). This project asks how to make that kind of search efficient when the annotations are produced by machine-learned models and change over time.

2. Research questions

- **RQ1. Index structures.** How should an inverted index represent model-produced labels and numeric scores so that queries combining keyword relevance, label filters and score ranges are answered with top-k efficiency, rather than by retrieving and post-filtering?
- **RQ2. Query processing.** Can dynamic pruning methods, such as WAND and block-max variants, be extended so that score ranges and label predicates prune early, and what is the efficiency and effectiveness trade-off against post-filtering and against separate per-label indexes?
- **RQ3. Maintenance under relabelling.** When a new model version changes a fraction of the labels, what is the cheapest way to keep the index correct: versioned postings, delta indexes merged in the background, or selective rebuilding? Can a cost model predict which strategy wins from the fraction and distribution of changed labels?
- **RQ4. Learned representations.** How do these techniques interact with learned sparse retrieval (for example SPLADE) and late-interaction dense retrieval (for example ColBERT) on short, noisy and multilingual social media text, where vocabulary and expansion terms differ from benchmark collections?

3. Theoretical perspectives

The project builds on three lines of work. **Efficient query evaluation over inverted indexes:** document-at-a-time processing with dynamic pruning (Broder et al., 2003; Ding and Suel, 2011) and its adaptation to learned sparse weights (Mallia et al., 2022). **Indexing beyond terms:** annotation-aware and range-capable indexes as developed in the group's own work (GYANI; range search over temporal and numerical expressions), which treat structured annotations as first-class index content. **Learned retrieval models:** learned sparse expansion (Formal et al., 2021) and late interaction (Khattab and Zaharia, 2020), whose efficiency depends heavily on posting-list statistics that differ for short social media texts. The working hypothesis is that annotation predicates are selective and skewed (most posts are neutral; few are toxic), so index layouts that co-locate label and score information with postings can let the engine skip most of the collection.

4. Methodological design

Systems. I would implement the methods in C++ on top of an established research engine such as PISA (Mallia et al., 2019), so that results are comparable with the literature and efficiency is measured on a realistic code base.

This is also how I would close my main gap: my production work is in Python, and this project would make me fluent in systems-level C++.

Data. (a) public social media corpora collected anew through official APIs within platform terms and under NTNU ethics approval, of the same kind as the four-platform policy corpus from my MSc research; (b) standard retrieval benchmarks such as MS MARCO and BEIR (Thakur et al., 2021) for effectiveness comparisons; (c) annotation layers produced by several model versions (a lexicon, a BiLSTM, fine-tuned transformers and an LLM annotator) to create realistic relabelling events for RQ3. Multilingual subsets (English, Urdu, Arabic) test RQ4.

Workloads and measurement. Query logs are rare for annotation-heavy analytics, so I would derive realistic workloads from analyst tasks (for example, monitoring reactions to policy events) and generate controlled variants with known selectivity. Efficiency is measured as mean and tail query latency, index size and update cost; effectiveness with standard measures (nDCG, recall at k) and with the exactness of filtered results against a brute-force baseline. Every comparison would report both efficiency and effectiveness, and results would be repeated across hardware settings, since modern hardware (SIMD decoding, multi-core, cache behaviour) changes which layout wins.

Validity. My MSc and manuscript work taught me that a label is only as good as the reference it is scored against. For RQ3 and RQ4, relabelling events will be characterised (how many labels change, and where), so efficiency results can be tied to the shape of the change rather than to one model pair.

5. Progress plan (three years of research, plus a fourth year of duty work if offered)

Period	Work	Output
Months 1 to 6	Courses in the IE doctoral programme; literature review; C++ proficiency on PISA; build the annotated corpora and the relabelling events; baseline post-filtering system.	Benchmark and workload release; review chapter
Months 7 to 15	RQ1 and RQ2: annotation-aware posting layouts and predicate-aware pruning; efficiency study against baselines.	Paper 1 (target SIGIR or CIKM)
Months 16 to 26	RQ3: versioned postings, delta indexes and selective rebuilding; a cost model for relabelling.	Paper 2 (target VLDB or SIGMOD)
Months 27 to 33	RQ4: learned sparse and late-interaction retrieval on noisy, multilingual social media text with annotation predicates.	Paper 3 (target TheWebConf or ECIR)
Months 34 to 36	Integration, open-source release of the code, thesis writing.	Thesis

Teaching assistance in Databases, Data Mining and Information Retrieval runs alongside, as described in the position. The outline is a starting point; I would refine the questions with the supervisor and the group in the first months.

References

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- Formal, T., Piwowarski, B. and Clinchant, S. (2021). SPLADE: Sparse lexical and expansion model for first stage ranking. SIGIR.
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- GYANI: An indexing infrastructure for knowledge-centric tasks (2018). CIKM. Scalable range search over temporal and numerical expressions (2024). ICTIR. (Databases and Search group, NTNU.)